Microsoft

AI infused mobile & desktop app development with .NET MAUI

David Ortinau | Product Manager
Beth Massi | Product Manager
Gerald Versluis | Engineer

Build intelligent apps, intelligently

.NET Agentic Apps

Infuse AI into your .NET apps

- Cloud
- Web
- Desktop
- Mobile
- Games
- IoT

Aspire

Build, test, and deploy seamlessly from code to cloud

Agentic DevOps

AI-powered agents operating as a member of your dev team

- Visual Studio
- Visual Studio Code
- GitHub
- Azure

Applications **Tools**

“AI is introducing the third user-interface paradigm in computing history, shifting to a new interaction mechanism where users tell the computer what they want, not how to do it — thus reversing the locus of control.”

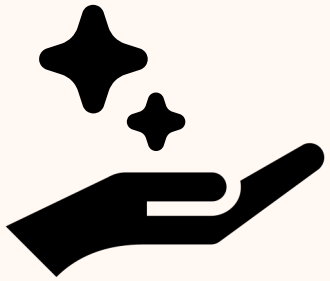
Jakob Nielsen



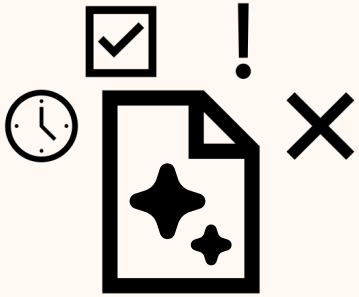
An AI-infused App

AI is changing applications, and how we make them

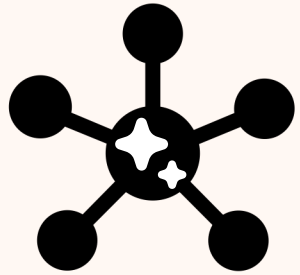
Changes in User Experience



Personalization



Context-awareness



Multimodal interaction

6 Design Principles for AI Applications

Design Responsibly

- Solve real user problems
- Minimize user harms
- Use human-centered design
- Expose or limit emergent behaviors
- Test & monitor for user harms

Design for Mental Models

- Orient the user to generative variability
- Teach effective use
- Understand the user's mental model
- Teach AI about the user

Design for Appropriate Trust & Reliance

- Help a user know when they should/should not rely on AI output
- Be clear on how well AI performs for a given task
- Provide rationales for outputs
- Use friction to prevent overreliance
- Encourage critical thinking
- Signify the role of the AI

Design Principles for Generative AI Applications
Justin D. Weisz, Jessica He, Michael Muller, Gabriela Hoefer, Rachel Miles, Werner Geyer International Conference on Human Factors in Computing Systems 2024

6 Design Principles for AI Applications

Design for Generative Variability

- Leverage multiple outputs
- Visualize the user's journey
- Enable curation & annotation
- Draw attention to differences or variations across outputs

Design for Co-Creation

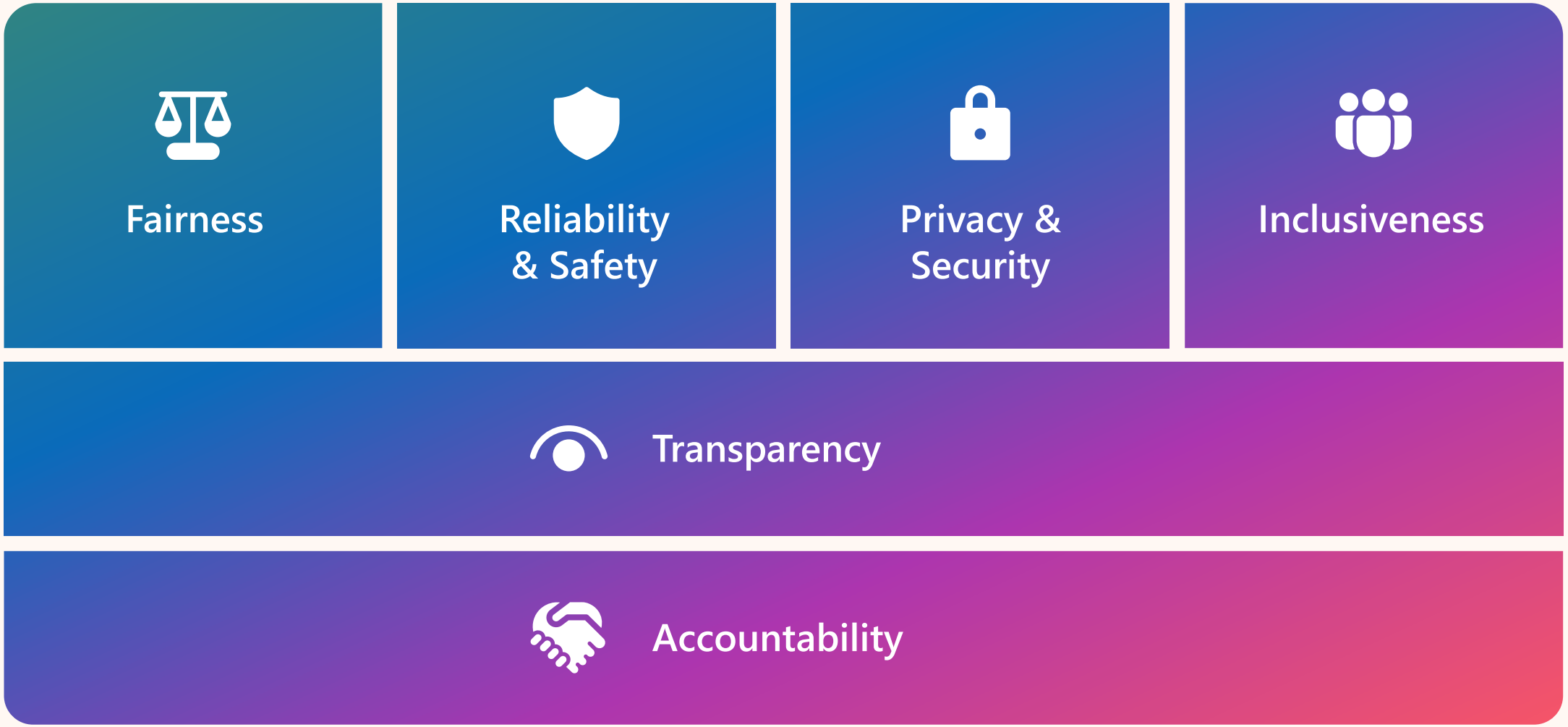
- Help the user craft effective outcome specifications
- Provide generic input parameters
- Provide controls relevant to the use case & technology
- Support co-editing of generated outputs

Design for Imperfection

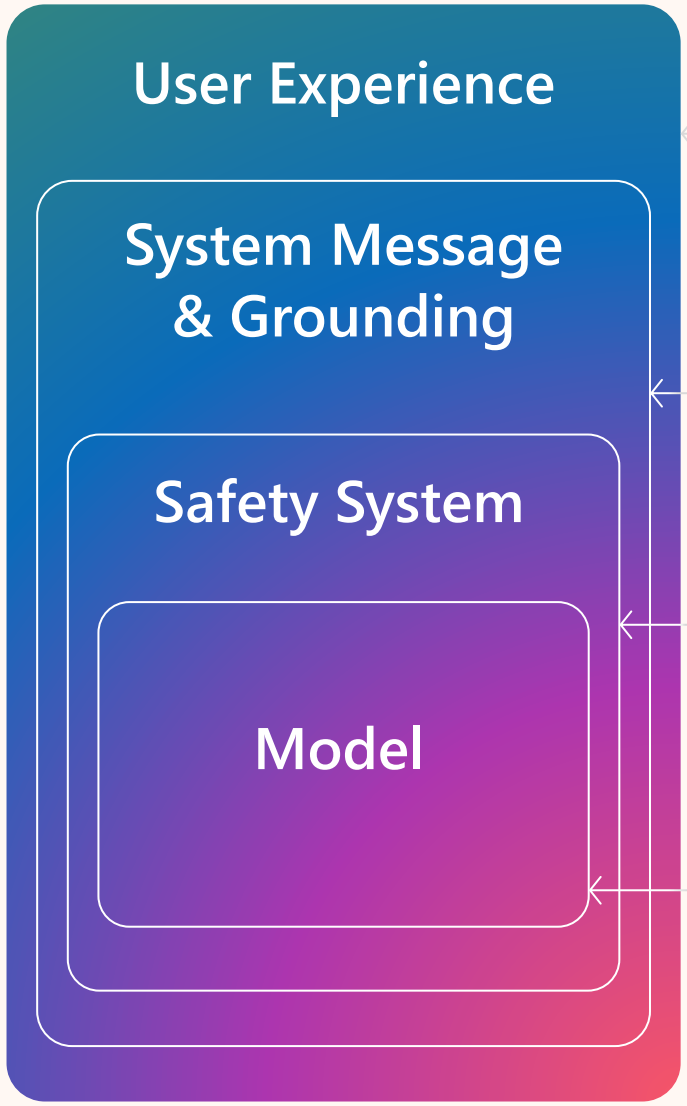
- Make uncertainty visible
- Evaluate outputs for quality
- Offer ways to improve outputs
- Provide feedback mechanisms

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Microsoft's AI Principles



Mitigation Layers



Design for responsible human-AI interaction

The HAX Toolkit is a set of practical tools to help create responsible and effective human-AI interactions, including guidelines, workbooks, design patterns, and a design library

Ground your model and steer its behavior

Recommended System Message Framework provides instructions, clear directives, background information relevant to the task, guidelines on tone and style of responses, and safety guidelines

Monitor and protect model inputs and outputs

Azure AI Content Safety detects and moderates harmful user-generated and AI-generated content in applications and services using text and image APIs

Choose the right model for your use case

Azure AI Foundry model catalog provides a broad selection of models and helpful tools to guide your choice

Resources for getting started

Microsoft's approach foundations

- 2024 Transparency Report
- Responsible AI Tools and Practices | Microsoft AI
- Responsible AI Standard v2
- Responsible AI Impact Assessment Template
- Azure AI to help you build more secure and trustworthy generative AI applications

Special topics

- Planning red teaming for large language models (LLMs)
- Microsoft's open automation framework to red team generative AI Systems
- Lessons from Red Teaming 100 Gen AI products whitepaper
- HAX Playbook for human-AI interaction design
- Customer Copyright Commitment Required Mitigations
- Accelerating Sustainability with AI: A Playbook

Learning modules

- Microsoft AI - Learning and Community Hub
- Embrace responsible AI principles and practices - Training
- Fundamentals of Responsible Generative AI – Training
- Responsible generative AI in AI Studio – Training
- AI Red Teaming Learning hub
- AI Literacy Starting Guide

Microsoft on the Issues

- How do we best govern AI?
- Our commitments to advance safe, secure, and trustworthy AI
- Microsoft announces new Copilot Copyright Commitment
- Combating AI deepfakes in elections through today's new tech accord
- Protecting the data of our commercial and public sector customers in the AI era
- Global Governance: Goals and Lessons for AI
- Innovating in line with the European Union's AI Act

Find resources like these and more at **aka.ms/RAI**

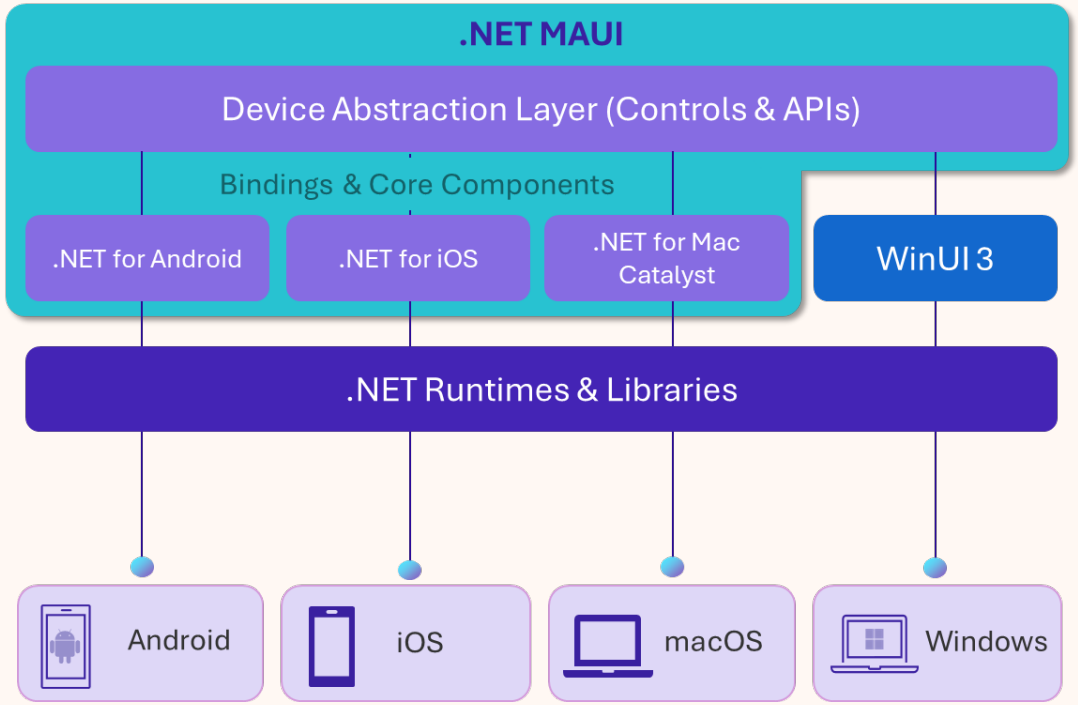
.NET MAUI is Infusion Ready

github.com/**dotnet/maui**

.NET MAUI

Build native, multi-platform apps all in one framework.

- Modern native client stack for .NET
- Single project, single codebase
- Native controls and styling
- Access over 60 common device APIs
- Access platform specific APIs if needed
- App store distribution and reach
- Productivity for Visual Studio and VS Code
- Platform is open source on GitHub
- Healthy ecosystem of controls and components



.NET MAUI NOW

+17% YoY Developers	**1.5k** Issues Resolved Since 9 GA	**1.2k** PRs Merged Since 9 GA
-26% Issues Created Last 4mo YoY	**4.7x** PRs Created Last 4mo YoY	**1.4x** PRs Merged Last 4mo YoY

One solution, everything you need

Context-awareness

Location
Calendar
Contacts
Time

Adaptive UI

App Theme Manager
Data binding
Visual State Manager
Templated layout
Adaptive Triggers

AI

Open AI
Microsoft.Extensions.AI
Semantic Kernel
Model Context Protocol
Tool calling
Vector search

Multimodal

Biometrics
Sensors
Text
Voice
Vision

Model Support

Azure AI Foundry
Open AI
Ollama
ONNX

UI Controls

ActivityIndicator
BlazorWebView
Border
BoxView
Button
CarouselView
CheckBox
CollectionView
ContentView
DatePicker
Editor
Ellipse
Entry
GraphicsView
HybridWebView
Image
ImageButton
IndicatorView
Label
Line
Map
Path
Picker
Polygon
Polyline
ProgressBar
RadioButton
Rectangle
RefreshView
RoundRectangle
ScrollView
SearchBar
SkiaSharp
Slider
Stepper
SwipeView
Switch
TimePicker
TwoPaneView
WebView

Microsoft.Extensions.AI

ChatClientServices.cs

```
_chatClient = new OpenAIClient(apiKey)
   .GetChatClient(model: model)
   .AsIChatClient();

_chatClient = new ChatClientBuilder(_chatClient)
   .Build();
```

MCP and Tool calling

ChatClientServices.cs

```
_chatClient = new OpenAIClient(apiKey)
  .GetChatClient(model: model)
  .AsIChatClient();

_chatClient = new ChatClientBuilder(_chatClient)
  .ConfigureOptions(options =>
  {
    options.Tools ??= [];
    options.Tools.Add(AIFunctionFactory.Create(_locationTools.IsNearby));
  })
  .UseFunctionInvocation()
  .Build();
```

Tool calling

LocationTools.cs

```
[Description("Checks if the user is near a location or business type")]
public async Task<string> IsNearby(
  [Description("Type of location (e.g. coffee shop, Target, grocer)")]
  string pointOfInterest,
  [Description("Distance threshold in meters (default: 100)")]
  double distanceThresholdMeters = 100)
{
  // use Map.APIs to get nearby locations
}
```

Bring it all together – Analyze & Prioritize Tasks

MainPageModel.cs

```
var apiResponse = await
chatClient.GetResponseAsync<PriorityTaskResult>(prompt);
if (apiResponse?.Result != null)
{
  if (!string.IsNullOrEmpty(apiResponse.Result.PersonalizedGreeting))
  {
    PersonalizedGreeting = apiResponse.Result.PersonalizedGreeting;
  }

  // do more
}
```

Plugin.Maui.Audio

VoicePageModel.cs

```
_recorder = _audioManager.CreateRecorder();
await _recorder.StartAsync();

_audioSource = await _recorder.StopAsync();

await using (var fileStream = File.Create(audioFilePath))
{
  var audioStream = _audioSource.GetAudioStream();
  await audioStream.CopyToAsync(fileStream);
}
```

MediaPicker

PhotoPageModel.cs

```
result = await MediaPicker.CapturePhotoAsync(
  new MediaPickerOptions
  {
    Title = "Take a photo"
  }
);

using Stream sourceStream = await result.OpenReadAsync();
using FileStream localFileStream = File.OpenWrite(ImagePath);
```

Project dependencies

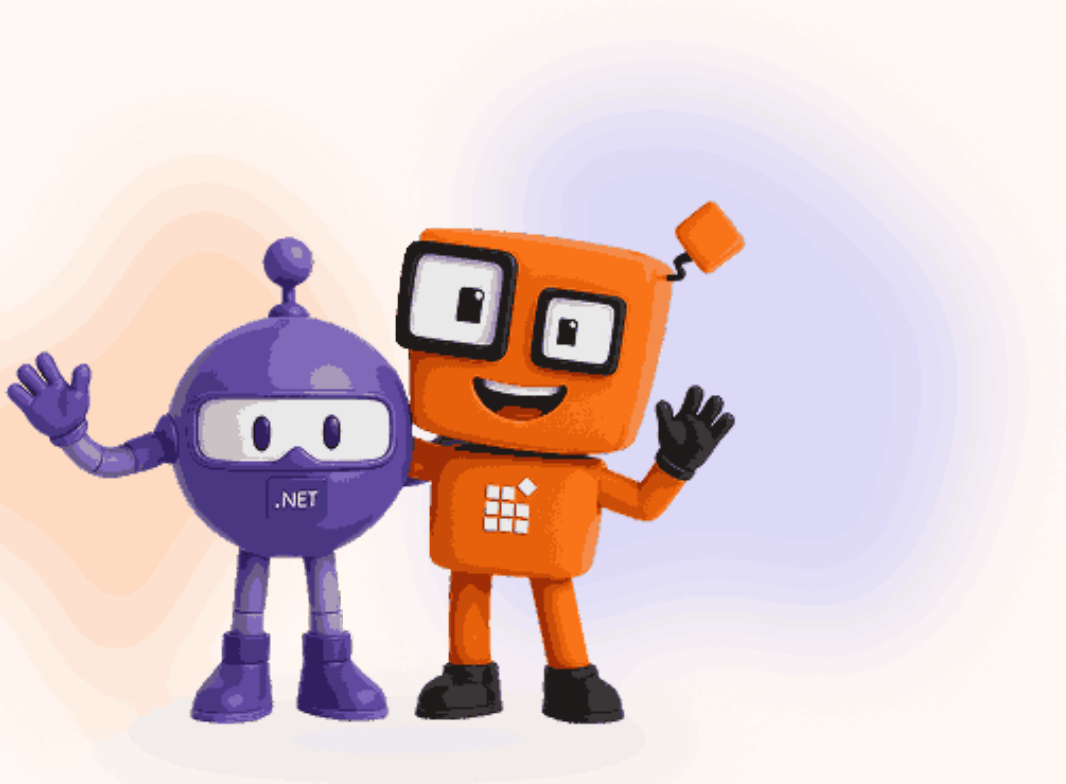
Libraries

- CommunityToolkit.Mvvm
- CommunityToolkit.Maui
- Microsoft.Extensions.AI
- Microsoft.Extensions.AI.OpenAI
- Microsoft.Maui.Controls
- Plugin.Maui.Audio
- Plugin.Maui.CalendarStore
- SkiaSharp

Syncfusion Toolkit for .NET MAUI

- BottomSheet
- CardView
- EffectsView
- Expander
- PullToRefresh
- SegmentedControl
- Shimmer
- TextInputLayout

Syncfusion + Microsoft: Empowering Developers Together





About Me

Uma Maheswari Chandrabose

Senior Product Manager, Essential Studio

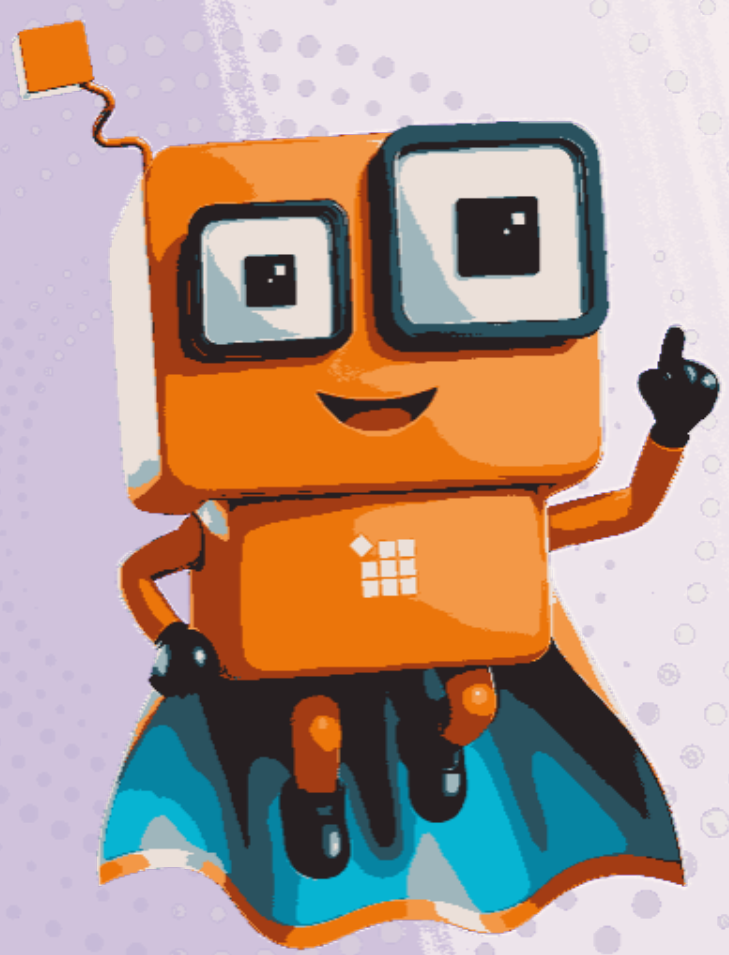
Syncfusion

https://www.linkedin.com/in/uma-maheswari-chandrabose

Syncfusion

- A comprehensive suite of 1900+ UI Controls across different frameworks

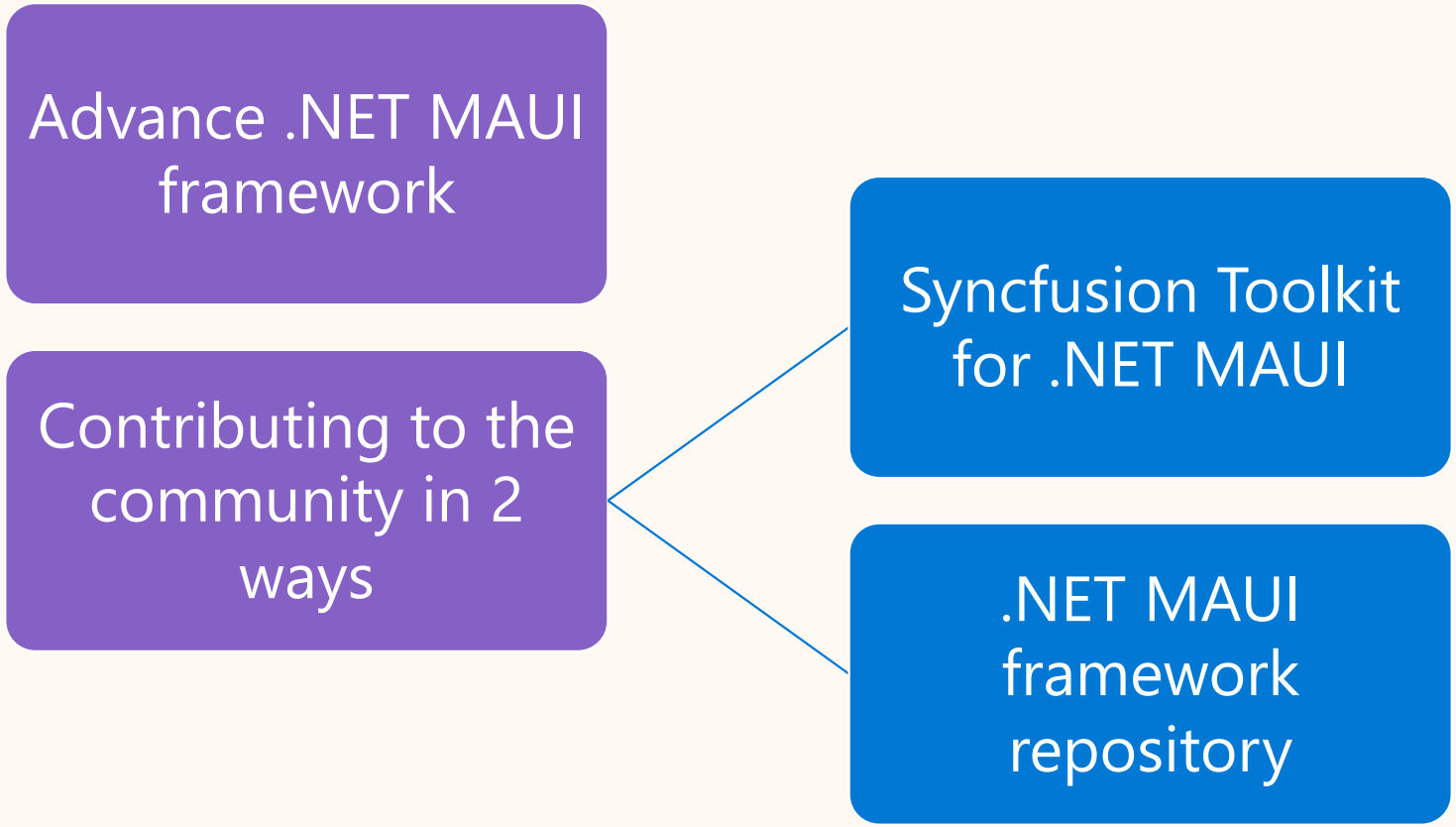
- Quickly build mobile, web, and desktop apps



How Syncfusion is Collaborating for Dev Community

Growing with .NET ecosystem

Microsoft Collaboration



Microsoft Collaboration

.NET MAUI Framework Repository

320 +

Merge Requests

100 +

PRs Under Review

65 %

of Community PRs

Impact of Collaboration



Speed up development – high quality and compatible



Get to know more about the community requirements.



Test our controls in beta version of .NET MAUI before the official release.



Huge spike in .NET MAUI Toolkit

New Updates to Syncfusion Toolkit for .NET MAUI

6 new controls for .NET MAUI Toolkit	1. Pickers – Picker, Date Picker, Time Picker, DateTime Picker 2. Progress bar – Linear, Circular
Optimized for AOT and Trimming	Your apps stay fast and light



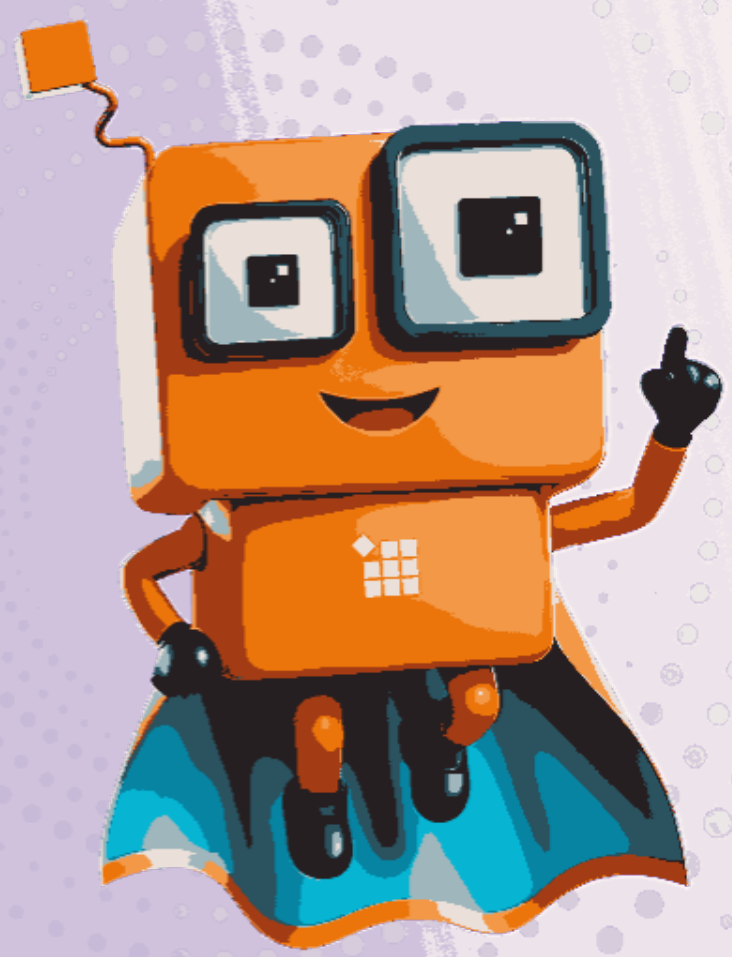
Final Note

Simplify development and deliver innovation with ease

AI-powered UI components

Thank you!

Do visit our website **syncfusion.com** for more details!



.NET MAUI Hybrid Apps

What are hybrid apps and why build them?

01 **Blend of native and web technology:**
UI is written using web technologies such as HTML, CSS, and JavaScript wrapped in a native app container.

02 **Access to native platform features and device hardware:**
leverage native platform features and device hardware that a web application cannot access easily or even at all.

03 **Can be mobile and/or desktop devices:**
run on multiple device platforms without the need for completely separate development.

04 **Reach via app store distribution:**
hybrid apps are native apps that can be downloaded and installed through the app stores.

05 **Code reuse across device platforms and web browsers:**
achieve greater developer productivity via code reuse across devices and web browsers.

Web UI and Native UI

Look and feel

Web UI typically looks the same across devices (controlled by CSS), while Native UI typically provides device-specific look & feel by default

UX

Native UI can provide more seamless user experiences tailored to the specific devices

Code reuse

Web UI can easily be reused in a web app

Skillset

Web UI requires knowledge of HTML, JavaScript, CSS, while Native UI (on the Microsoft stack) requires knowledge of XAML

.NET MAUI Hybrid Apps

Hybrid apps with Blazor

- Solution template for Blazor Hybrid+Web
- Tight inner loop with Copilot and Hot Reload
- Built-in input controls, validation, forms, auth



Hybrid apps with other JavaScript frontends

- Reuse Angular, React, etc. investments in MAUI native desktop and mobile apps
- Provides C# to JavaScript interop
- New hooks for intercepting web requests in .NET 10 (P5)

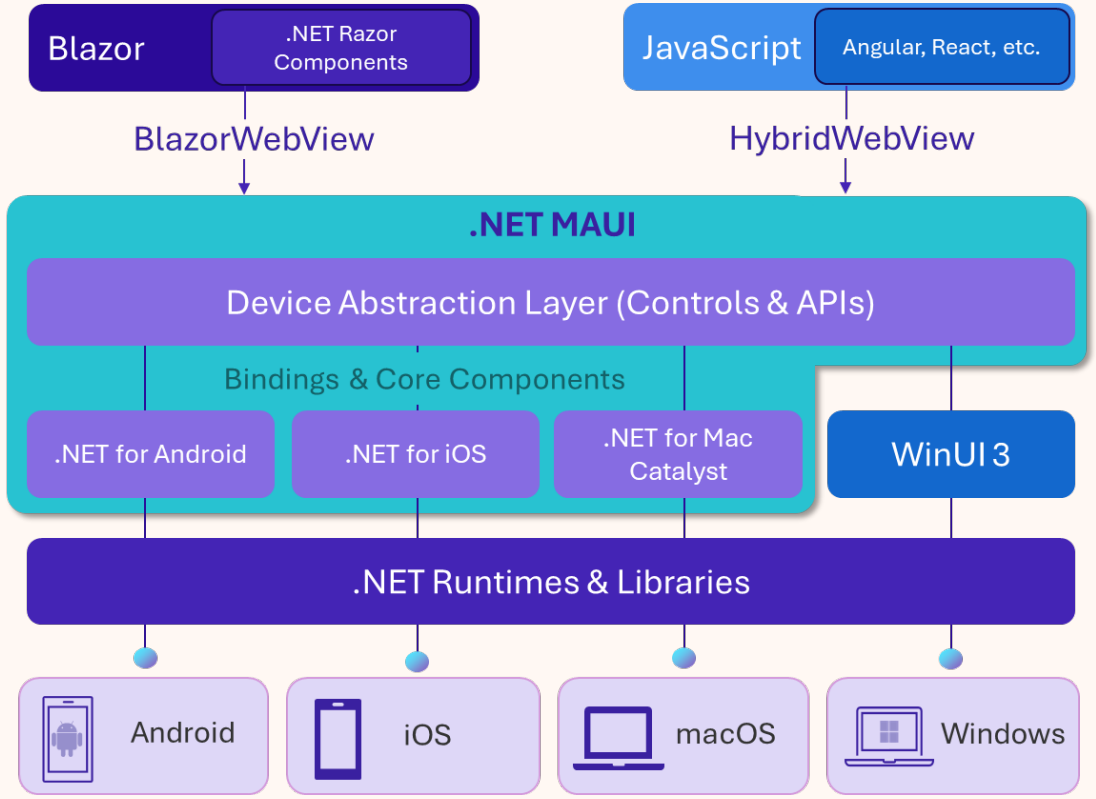


github.com/**dotnet/maui**

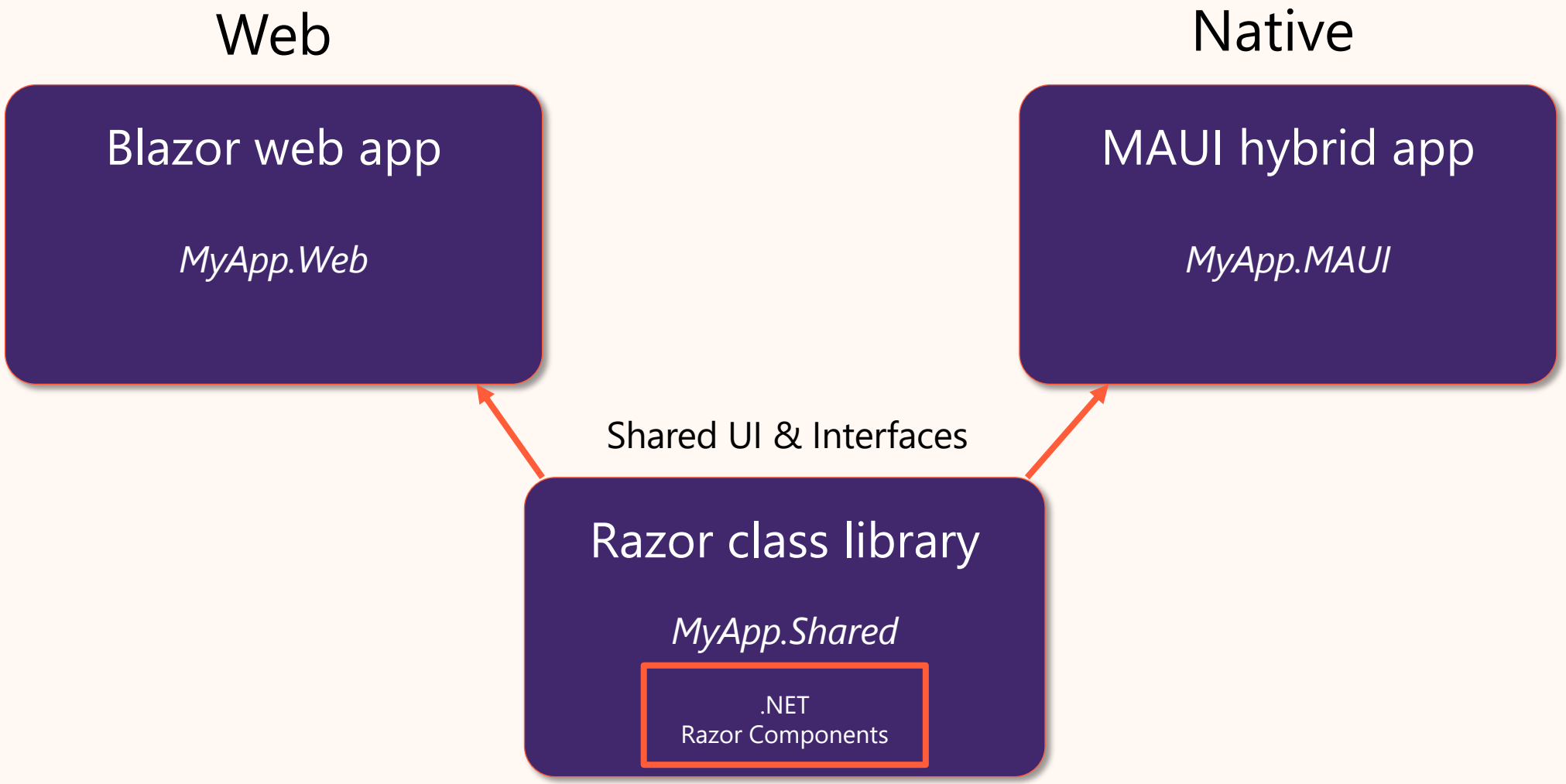
.NET MAUI Hybrid Apps

Build native, multi-platform apps with web UI.

- Native app container & embedded controls
- Fully native web UI through platform WebView
- No internet required; web assets packaged with app
- Multi-platform with MAUI abstractions
- App store distribution and reach
- Easily reuse web UI across MAUI native and Blazor web apps with new Solution template
- Support for Blazor and other JavaScript frameworks like Angular, React, Vue, etc.



Reusing UI across native and web with Blazor



DEMO: .NET MAUI Blazor Hybrid and Web



Agentic DevOps

Evolving DevOps

DevOps

"Union of **people, process, and technology** to enable continuous delivery of value to end users."

DevSecOps

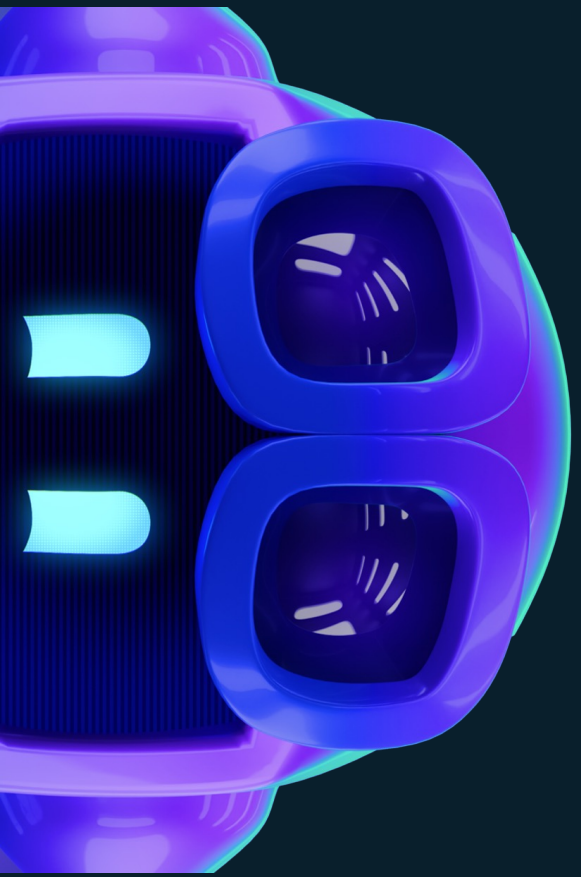
"Union of **people, process, and technology with security** as a shared responsibility to enable continuous delivery of value to end users."

Agentic DevOps

AI-powered agents operating as a member of your dev team to **automate, optimize, and accelerate every stage** of the software development lifecycle.

GitHub Copilot is built for developer experience

Automate the mundane and focus on what matters

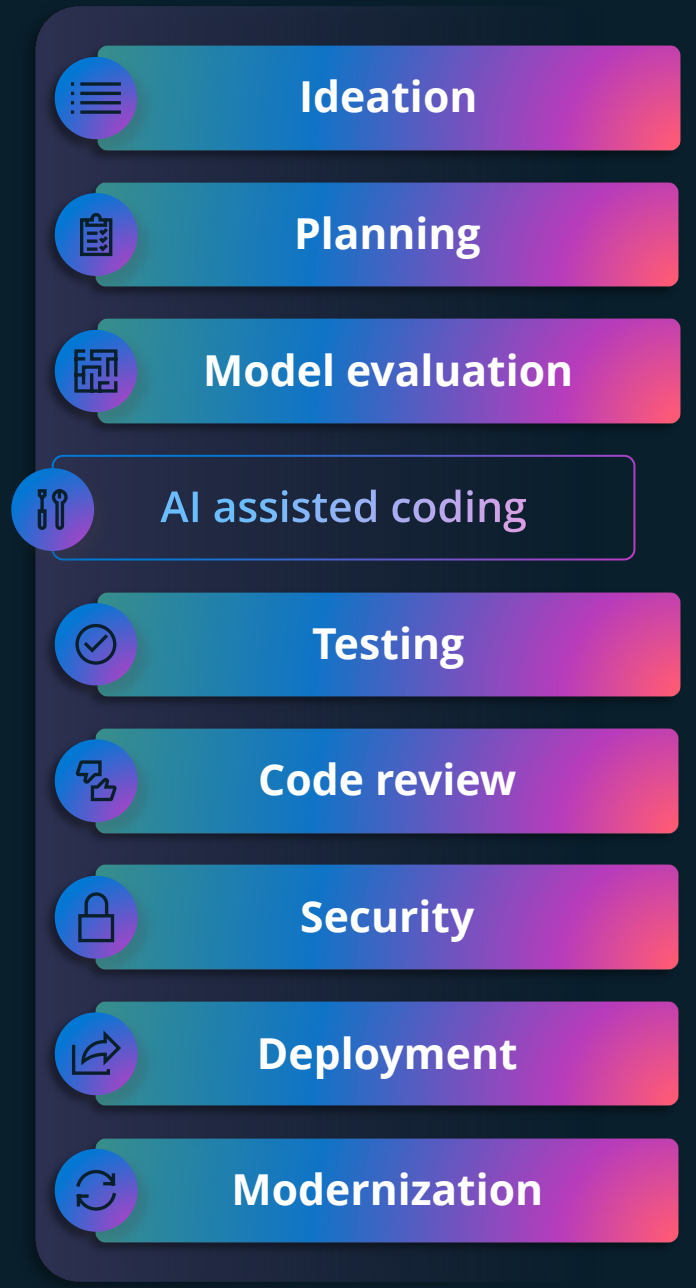


Experiences

- Sense making (summarization, code understanding)
- Writing code
- Writing tests, documentation
- Debugging code
- Reviewing code
- Fixing bugs and vulnerabilities
- Upgrading dependencies
- Code modernization, refactoring
- Updating stakeholders

The next evolution of GitHub Copilot

New agents help solve real-world problems throughout the software lifecycle

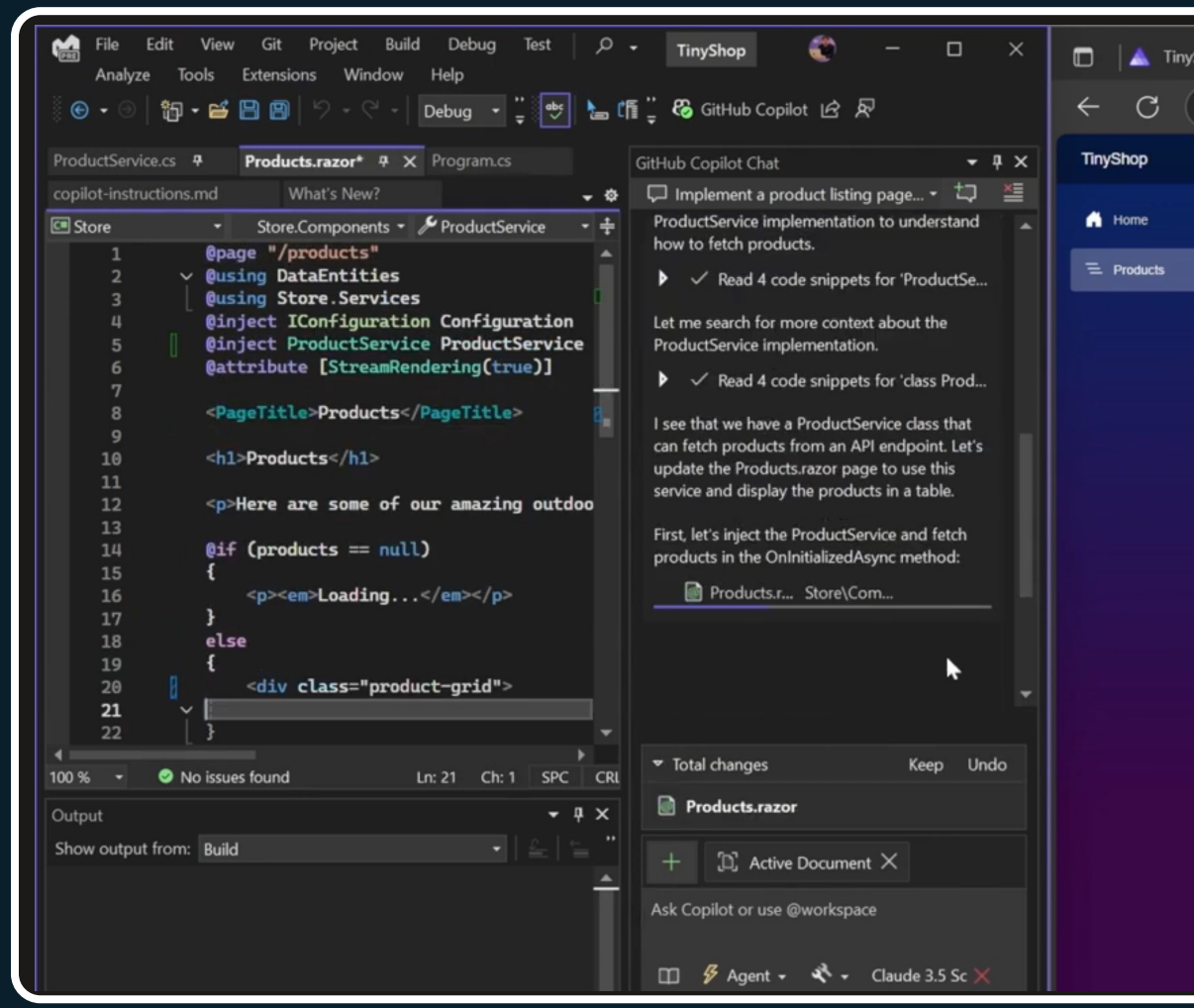


GitHub Copilot

Agent Mode

Supercharge the AI-assisted coding experience directly in Visual Studio

- Expand your AI capabilities beyond autocomplete by assigning multi-step coding tasks directly from your IDE
- Leverage an AI-powered peer programmer that can build new features, refactor legacy code, and heal itself when things break
- Analyze a codebase, edit files, run tests, fix errors, and suggest terminal commands—all from a single prompt
- Use agent mode in the IDE of your choice with support for Visual Studio and Visual Studio Code + more

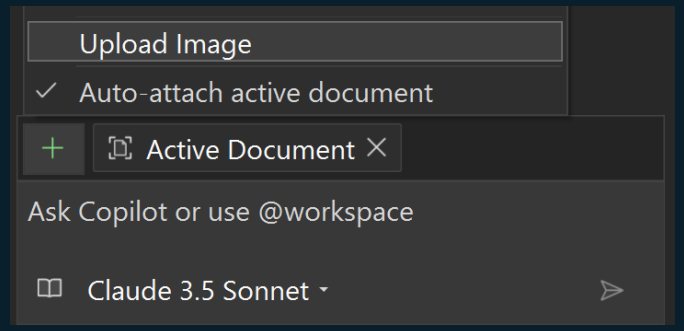


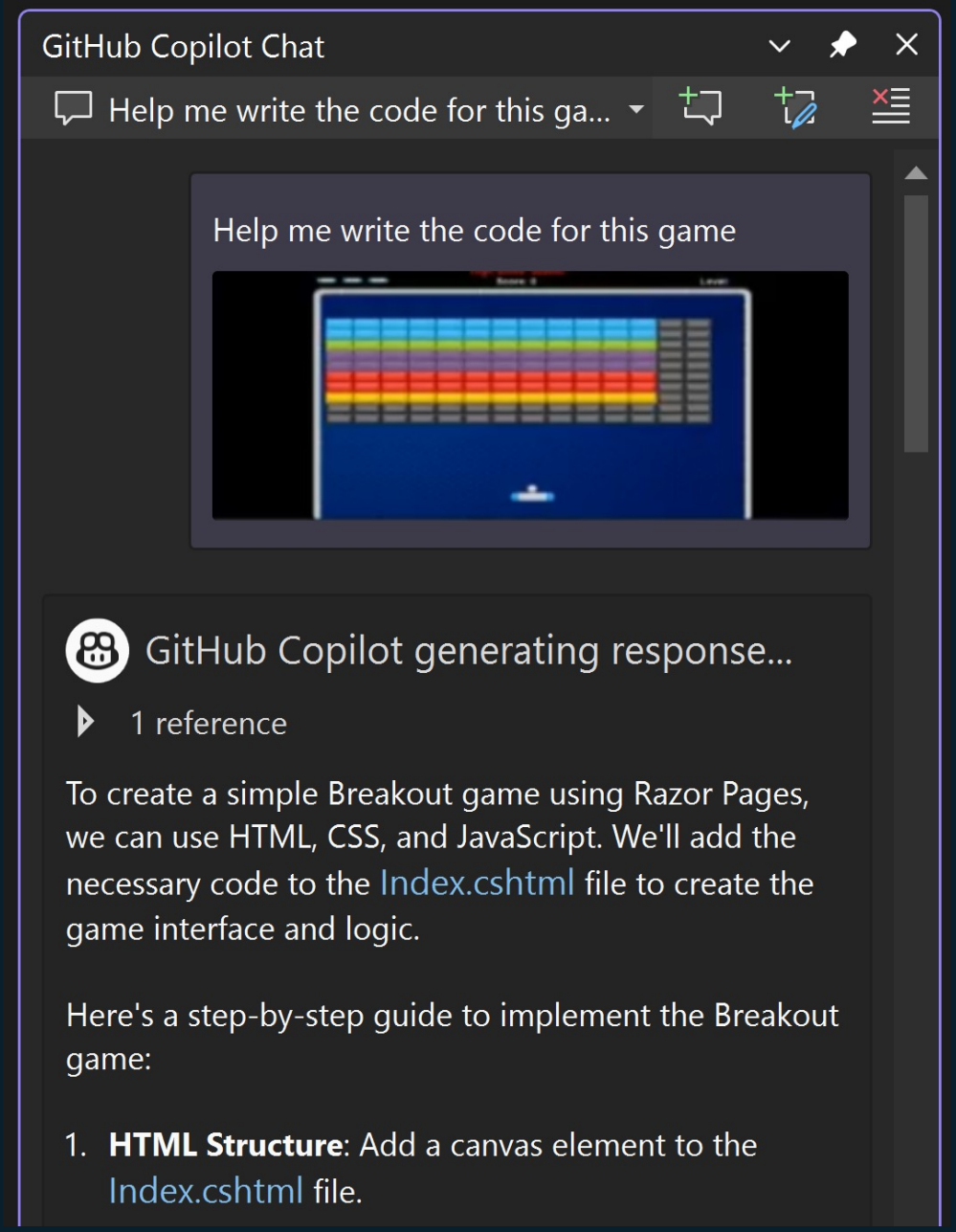
GitHub Copilot

Copilot Vision

Illustrate your context with images instead of text in Visual Studio

- Copilot interprets UI designs and specs, redlines and annotations, project architecture diagrams, data schemas, even napkin sketches
- Together with hot reload and live preview, watch your UI designs come to life
- Provides the latest multi-modal LLM improvements in Visual Studio



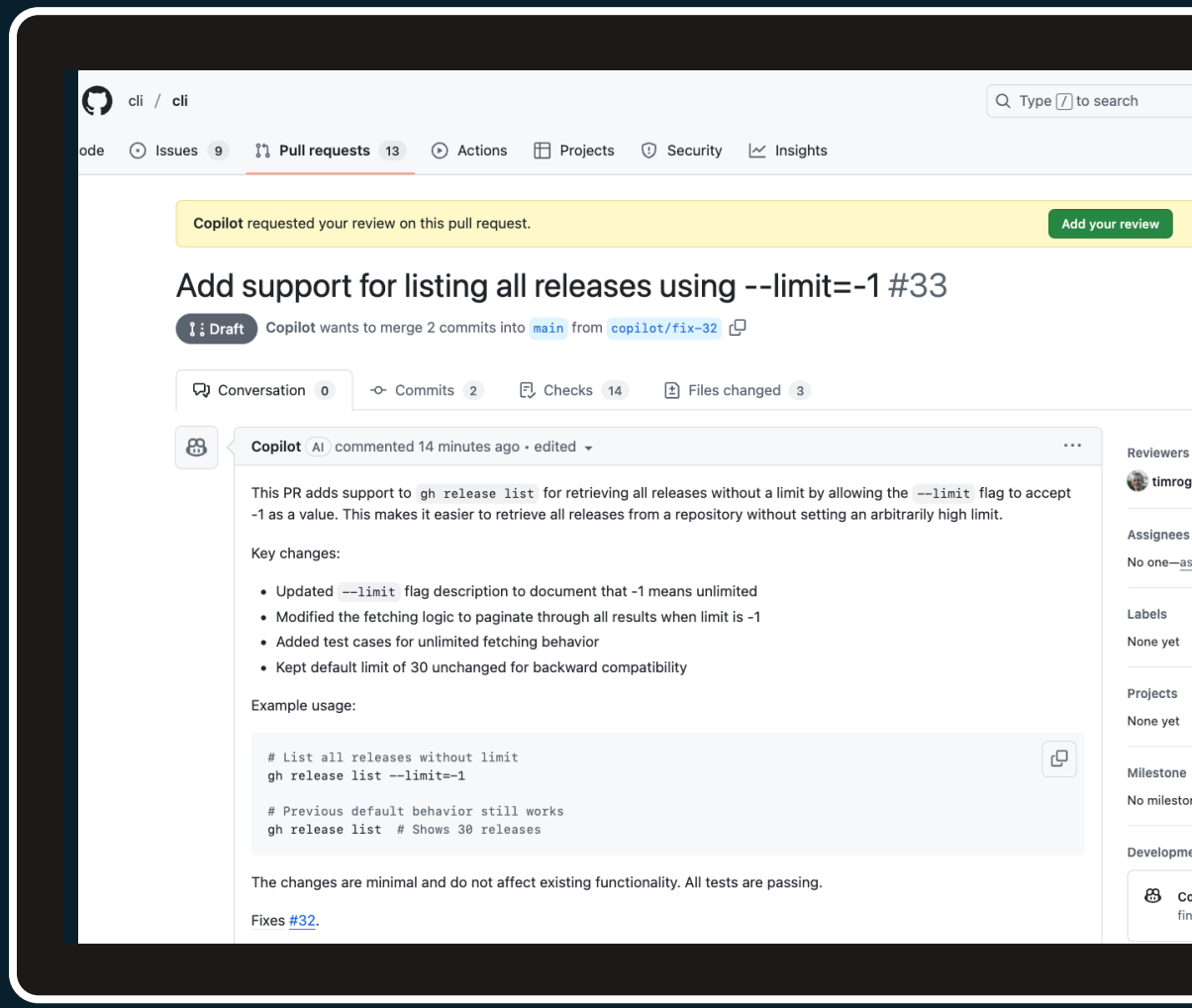


GitHub Copilot

Coding Agent

Agentic capability in GitHub to enhance team productivity like an AI teammate

- Assign issues to GitHub Copilot and get back pull requests, automatically validated with tests and linters
- Stay in control by working with Copilot just like any other developer and iterating through pull request reviews
- Tailor Copilot to your needs by customizing its dev environment, adding custom instructions, and providing external context through MCP servers



Demo: Intelligent Productivity

Copilot Agent, Vision, and Live Preview





.NET Aspire



Code-first control



Modular & extensible



Observability from the start

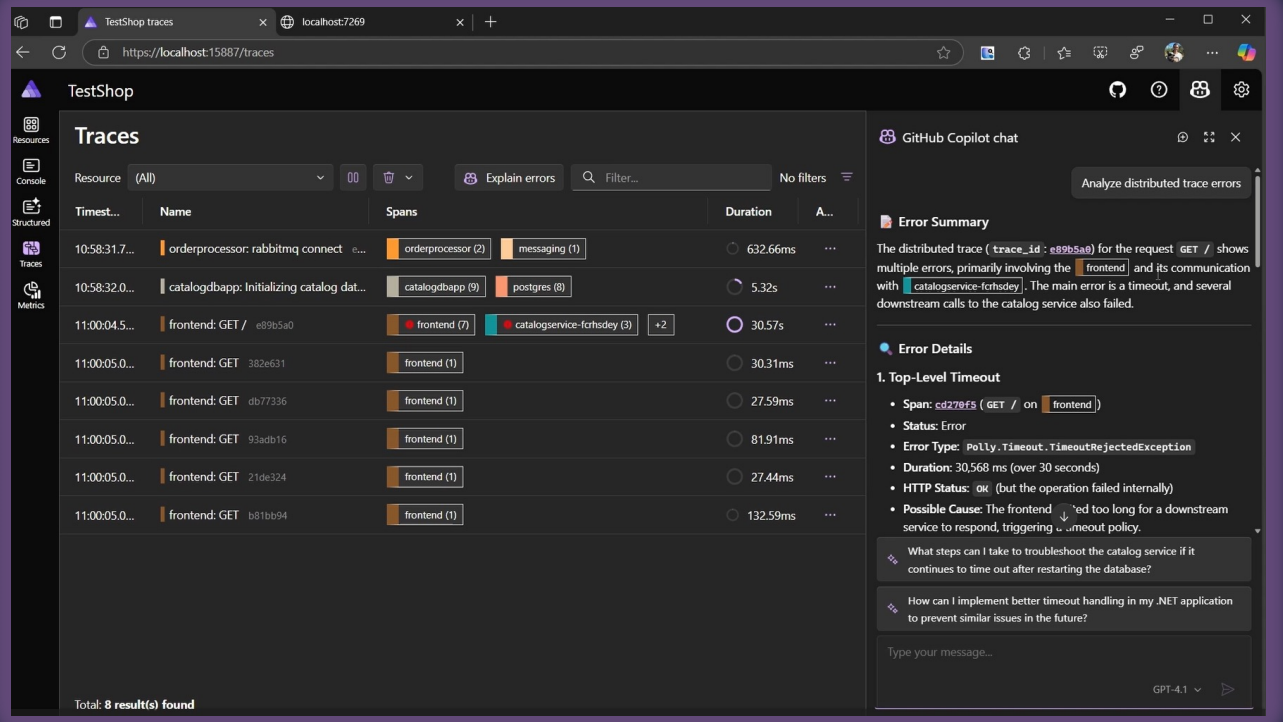


Flexible deployments

Aspire with Copilot

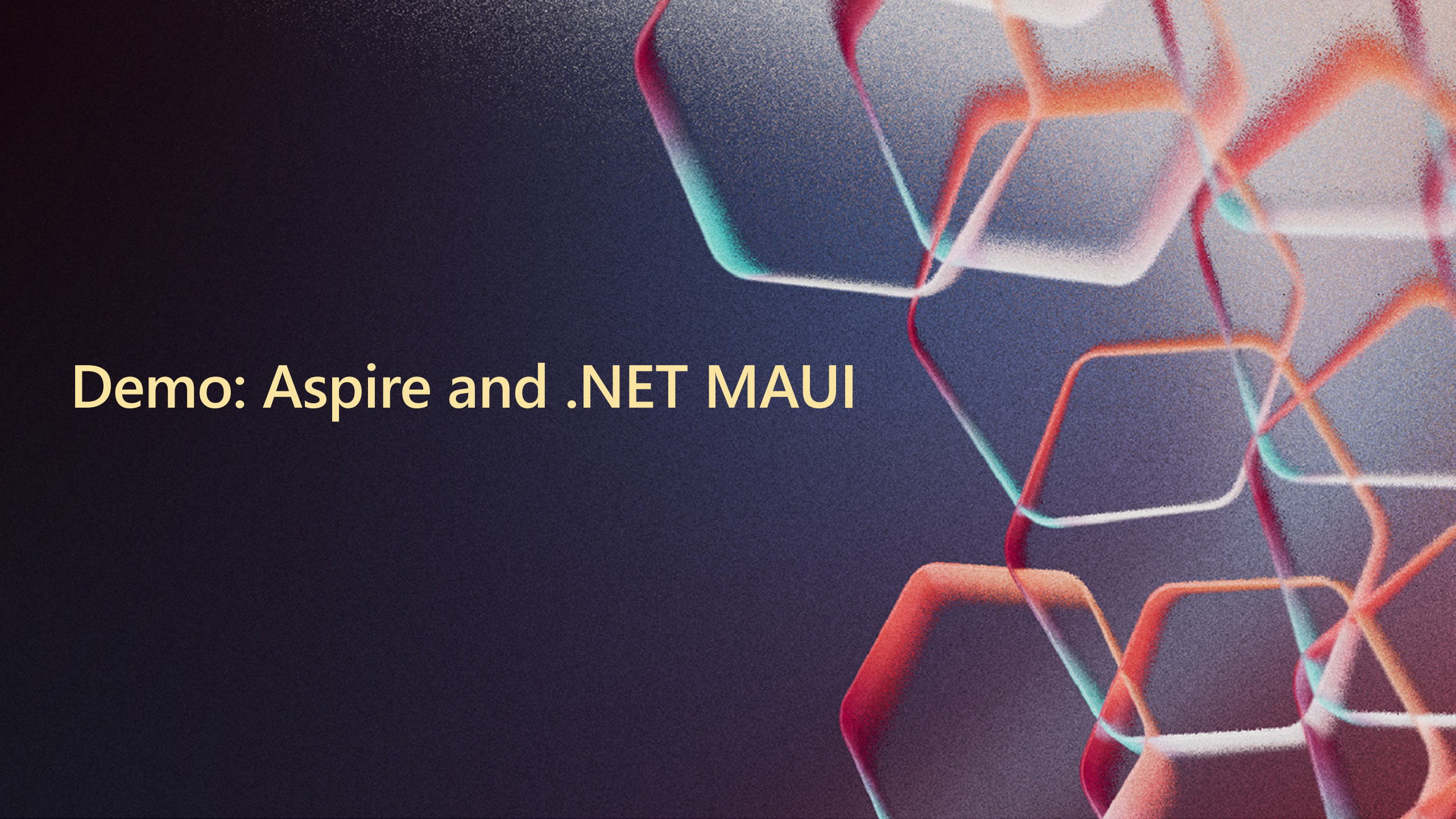
AI-assisted experience directly in the Aspire developer dashboard

- Have Copilot identify resource issues, find bottlenecks, and help with spotting problems at development time
- Ask Copilot about specific traces and find root causes of errors
- Works seamlessly with the rest of your Copilot agents in Visual Studio





Demo: Aspire and .NET MAUI



Recap

Learn more

Microsoft AI Principles
aka.ms/RAI

.NET MAUI
github.com/dotnet/maui

Microsoft.Extensions.AI
github.com/dotnet/extensions

Syncfusion Toolkit
github.com/

.NET MAUI Hybrid Apps
aka.ms/maui-hybrid-docs

.NET MAUI Hybrid Lab
aka.ms/build-lab305

Related sessions

DEM520 • Powering local AI development with .NET Aspire
May 19 11:35-11:50 AM Pacific

COMM443 • Table Talks: Building multiplatform apps with .NET MAUI
May 19 4:00-5:00 PM Pacific

COMM411 • Conversations: Let's Talk .NET Aspire
May 20 11:45-12:30 PM Pacific

BRK101 • The Future of .NET App Modernization Streamlined with AI
May 20 3:00-4:00 PM Pacific

BRK115 • Python Meets .NET: Building AI Solutions with Combined Strengths
May 20 4:30-5:30 PM Pacific

BRK106 • Elevating Development with .NET Aspire: AI, Cloud, and Beyond
May 21 2:00-3:00 PM Pacific

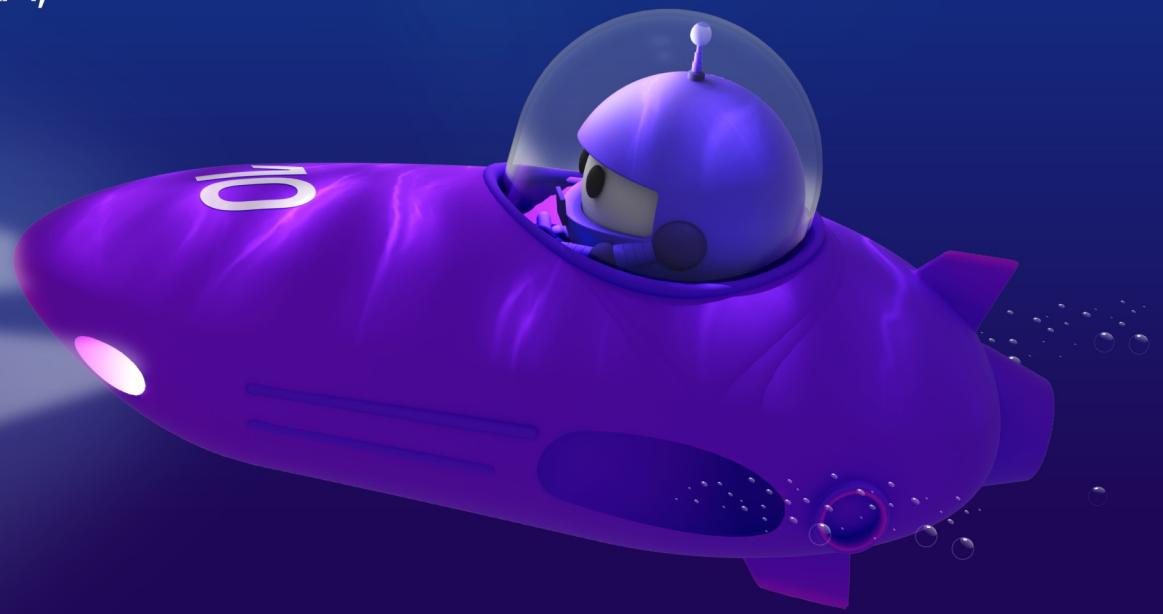
BRK123 • AI infused mobile & desktop app development with .NET MAUI
May 22 10:00-11:00 AM Pacific

Save the date

.NET Conf 2025

November 11-13

Official .NET 10 GA launch.
Join the 24-hr live broadcast with speakers, Q&A, games, prizes, and digital swag.



www.dotnetconf.net

Microsoft

Thank You!

David Ortinau | github.com/davidortinau
Beth Massi | github.com/BethMassi
Gerald Versluis | github.com/jfversluis

